Project Completion Report

Hydra Point-of-Sale: Accept ADA, CNTs in Retail & Invoicing

Field	Value
Project Name	Hydra Point-of-Sale: Accept ADA, CNTs in Retail & Invoicing
Project Number	1400065
Challenge	F14: Cardano Open: Developers
Project Manager	Rahul Konudula
Project Start Date	November 24, 2025
Project Completion Date	June 10, 2026
Repository	Trivolve-Tech/Hydra-Point-of-Sale-Accept-ADA-CNTs-in-Retail-Invoicing
Documentation site	trivolve-tech.github.io/Hydra-Point-of-Sale-…

1. Deliverables

Hydra Point-of-Sale delivered a **self-hosted, open-source point-of-sale + invoicing toolkit** that accepts **ADA and Cardano Native Tokens (CNTs)** with **Hydra Layer-2 settlement on Cardano mainnet** — real ADA, real Hydra heads, real CIP-30 wallets — rolled out for live retail payments at the **Sova Bar (Da Nang) launch event**. The system ships a single Next.js application that serves both a merchant dashboard (`/`) and a customer payment page (`/customer`), a per-customer Hydra-head orchestrator, and a sanitised mainnet docker-compose stack. **L1 commits are non-custodial** (CIP-30 wallet-signed); **L2 spends are server-signed** with a per-customer ed25519 key, AES-256-GCM encrypted at rest. Architecture in `docs/architecture/hybrid-custody.md`.

Output	Link
Documentation site (single URL)	https://trivolve-tech.github.io/Hydra-Point-of-Sale-Accept-ADA-CNTs-in-Retail-Invoicing/
Source repository	GitHub
Releases (downloadable, installable)	v1.0.0 · v2.0.0 — hybrid-custody mainnet rewrite
Mainnet deployment + ops	mainnet-deployment · key-custody · incident-playbook · deposit-period-tuning
Launch-event feedback (4 testimonials)	docs/FEEDBACK.md
In-app demo videos	docs/demo-videos/
Community event photos/videos	Google Drive folder
Specification PDFs (in-repo)	Requirements · Technical architecture

Components: `merchant-pos/` (Next.js 15 app + APIs + per-customer head orchestrator under `src/server/`), `docker/` (mainnet + per-head compose stacks), `hydra-config/`, `scripts/` (`derive-merchant-addresses.ts`, `reconcile.ts`).

Open source: Yes — **MIT** (LICENSE). **On-chain:** Cardano mainnet via Blockfrost; `hydra-node` 1.3.0. Hydra heads are per-`(merchant, customer)` operator-instantiated, so no canonical contract address — each deployment opens its own heads. **Testing:** testnet pilot on Preprod + Preview (v0.2.x), mainnet bring-up verified end-to-end via `docker compose -f docker/docker-compose.mainnet.yml`, plus `scripts/reconcile.ts` cross-checks intent state vs `hydra-node` `/snapshot/utxo`. **Live validation:** Sova Bar (Da Nang) launch drove real retail payments through the v2.0 stack. **User feedback:** four attendee testimonials captured in `docs/FEEDBACK.md`, headlined by *"We love it — it is processing payments in sub-seconds. This is the best crypto PoS system I've used so far."* — **Apparao, Partner, Sova Bar**.

2. Usage

Live deployment: Sova Bar (Da Nang) accepts ADA + CNTs from real retail customers on Cardano mainnet using v2.0.0. **Self-hostable:** any operator runs `docker compose --env-file docker/.env.mainnet -f docker/docker-compose.mainnet.yml up -d` and follows `docs/ops/mainnet-deployment.md`. Customers connect any CIP-30 wallet (Vespr, Eternl, Lace, Nami, Flint), see a six-step commit progress stepper with live ETA, then pay in-head at sub-second perceived latency. **Key actions completed: v1.0.0** mainnet baseline (2026-05-24) and **v2.0.0** hybrid-custody mainnet rewrite (2026-06-01) shipped to the public Releases page; docs site live on GitHub Pages; the Sova Bar launch event captured on photos/videos in the linked Drive folder. **Evidence of engagement:** Releases, Pages, FEEDBACK.md, community media.

3. Impact

Before this project there was **no self-hostable, open-source ADA + CNT PoS using Hydra L2 on Cardano mainnet**. v2.0 ships one — MIT-licensed, with a **better retail-customer experience** (smoother CIP-30 flow, six-step progress stepper, sub-second perceived L2 settlement) and a **better business experience** (single Next.js app, per-customer head orchestration, hybrid-custody model, sanitised mainnet env template, ops playbooks). **Performance / reliability:** in-head L2 settlement is sub-second in practice (cite Apparao's testimonial); L1 commit/close costs are paid **once per head**, not per payment; `--deposit-period` was raised from 60 s to 300 s after observing failures at 60 s on mainnet; L2 signing uses `csl.FixedTransaction.sign_and_add_vkey_signature` so the original body bytes (and body hash) are preserved; `HPOS_KEY_SECRET`, `WALLET_SEED_PHRASE`, `BLOCKFROST_KEY`, and `SITE_PASSWORD_MERCHANT` all fail-closed with no defaults. **Cardano ecosystem benefit:** the first open-source ADA + CNT PoS on Hydra L2 + mainnet; the per-head docker orchestrator, sanitised env templates, and ops playbooks lower the operational barrier for other Cardano builders. **Recognition:** partnership with Sova Bar (Da Nang) for live retail payments; four attendee testimonials (Apparao — venue partner; Bella, Charan, Maria — retail customers) documented in `docs/FEEDBACK.md`.

4. Sustainability

Ongoing, public, MIT-licensed. Maintenance: GitHub issues/PRs on the canonical repo; SemVer-tracked `CHANGELOG.md`; tag-triggered `release.yml` workflow; GH Pages auto-deploy on docs change; operational knowledge kept in-tree under `docs/ops/`. **Revenue model:** the toolkit is free and MIT — operators bear their own hosting + Blockfrost costs and Cardano L1 fees for head open/close. No protocol fee, no token, no central treasury. Continued development (broader CNT coverage, multi-merchant orchestration, mobile-first PWA polish, further hardening per the incident playbook) is pursued via follow-on Catalyst funding and bespoke integration engagements with Cardano-aligned venues. **Permanent storage + forking:** code is public on GitHub (MIT, fork-friendly); release artefacts attached to each tagged GitHub Release; the docs site rebuilds from the cloned repo. To fork: clone, follow the README quick start → `docs/ops/mainnet-deployment.md`; required values are documented inline in the sanitised `.env.mainnet.example`. No proprietary services are required to build or run the stack.

Project Completion Video (PCV): the final community launch was held at Sova Bar (Da Nang). Photos + videos: Google Drive folder. In-app product demos: `docs/demo-videos/`.